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Reframing pre-service teacher competencies in the age of AI-mediated CLIL instruction: a theoretical review

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Background: Artificial intelligence (AI) is reshaping the landscape of education, particularly in bilingual and subject-integrated contexts such as Content and Language Integrated Learning (CLIL). Yet, pre-service teacher education often lacks explicit guidance on how to embed AI into CLIL pedagogy in ways that are pedagogically meaningful, ethically responsible, and culturally inclusive. In particular, limited attention has been given to how AI mediates pedagogical processes and learner–teacher interactions in CLIL contexts.
Objectives: This study conducts a theoretical review to address three guiding questions: (1) What competencies for AI-mediated CLIL instruction are identified in existing literature? (2) How do established frameworks CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics interact in this emerging field? (3) How can these insights be synthesized into a new competency framework for pre-service teacher education?
Methods: This study reviewed interdisciplinary literature (2010–2025) from Web of Science, Scopus, and ERIC. A three-stage qualitative coding process (open, axial, selective) was applied by two independent coders (Cohen's $\kappa = 0.82$) to identify key competency domains across frameworks. Findings were synthesized to develop the CAIIL-TC framework, incorporating comparative insights from multiple regions.
Results: The CAIIL-TC framework specifies five domains: AI Content Evaluation, Generative Input Design, Contextual Re-Authoring, Ethical and Reflective Use, and Collaborative Mediation. These domains extend beyond existing models by directly addressing AI-mediated teaching tasks.
Conclusion: The CAIIL-TC framework provides a structured foundation for integrating AI into CLIL teacher preparation. It offers guidance for curriculum design, accreditation standards, and professional development, while underscoring the need for future empirical validation through classroom studies, longitudinal research, and cross-cultural comparisons.

KEYWORDS
AI literacy, artificial intelligence in education, bilingual instruction, CLIL pedagogy, pre-service teacher education, TPACK framework

1 Introduction

In recent years, the rapid development of artificial intelligence (AI) technologies has begun to reshape the educational landscape. From intelligent tutoring systems and automated feedback tools to large language models (LLMs) capable of content generation and translation, AI is becoming a transformative force across all levels of

education (Hossain et al., 2025; Hu-Au and Lee, 2018; Luckin et al., 2016; Son et al., 2025; Zawacki-Richter et al., 2019). Parallel to this trend, Content and Language Integrated Learning (CLIL) has gained traction as a pedagogical approach that fosters both disciplinary understanding and language proficiency through dual-focused instruction (Coyle et al., 2010; Dalton-Puffer, 2011; Llinares et al., 2012; Lyster, 2017; Pérez Cañado, 2018; Pérez-Cañado, 2011; Pérez-Vidal, 2009). As AI becomes increasingly embedded in educational practice, the intersection of AI and CLIL introduces new pedagogical possibilities (Chien et al., 2024; Lai, 2024) as well as critical challenges, particularly for pre-service teachers (Garamyan et al., 2025).

CLIL's pedagogical principles, centered around the 4Cs framework of Content, Communication, Cognition, and Culture, require teachers to simultaneously facilitate subject-specific knowledge, second language acquisition, higher-order thinking, and intercultural competence (Coyle et al., 2010). These demands already place a high cognitive and instructional load on educators, particularly in balancing language and content integration (Cammarata and Tedick, 2012). The integration of AI into CLIL environments further complicates this landscape, necessitating that teachers develop new competencies in technology use, ethical reasoning, content adaptation, and AI-informed pedagogical design (UNESCO, 2021; Zawacki-Richter et al., 2019).

However, current teacher education programs often lag behind these shifting demands. Many pre-service teachers receive limited exposure to AI tools or lack the digital literacy needed to evaluate and responsibly implement AI-generated materials (Alexandrowicz, 2024). At the same time, the affordances of AI, such as personalized learning, multimodal scaffolding, and adaptive feedback, are often underutilized in CLIL contexts, where the interplay between language and content requires nuanced, context-aware instruction (Son, Ružić, and Philpott, 2025). Without adequate training, teachers risk either rejecting potentially transformative tools or applying them in ways that undermine pedagogical coherence.

The need for reconceptualizing teacher competencies is further compounded by regional differences. In Europe, where CLIL has a stronger institutional presence, teacher education programs have begun incorporating digital pedagogy and media literacy into training frameworks (Costa and D'Angelo, 2011; Eurydice, 2017; Finnish Government, 2021; Marsh et al., 2018; Vuorikari et al., 2022). In contrast, Asian systems, such as Taiwan's national bilingual education initiative (Taiwan Today, 2025; Teng, 2024), and Japan's AI education framework (Ministry of Education, Culture, Sports, Science and Technology – Japan, 2022), face more structural and cultural barriers in equipping pre-service teachers to navigate AI-integrated CLIL classrooms. Moreover, ethical concerns, such as academic integrity, data privacy, algorithmic bias, and teacher displacement, are seldom addressed in pre-service curricula (Floridi et al., 2018).

Against this backdrop, this theoretical review aims to synthesize international scholarship on CLIL, AI integration, and teacher education. Drawing upon established frameworks, including TPACK (Koehler et al., 2013), CLIL's 4Cs (Coyle et al., 2010), AI literacy competencies (UNESCO, 2021), and professional ethics in digital education (Alexandrowicz, 2024), this review identifies the evolving competencies required of pre-service CLIL teachers in AI-mediated contexts. It proposes a multi-dimensional competency model to guide teacher preparation programs and offers regionally responsive recommendations, particularly for Europe and Asia. Through this synthesis, the review seeks to support the development of future-ready educators who can leverage AI for equitable, cognitively rich, and linguistically immersive instruction.

2 Theoretical framework

Teacher education in the age of AI-integrated CLIL requires an expanded conceptual foundation that goes beyond single-framework approaches. This section reviews four influential models: CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics, not in isolation, but through their intersections. While each framework has been widely used in education research, their integration provides a richer perspective for conceptualizing teacher competencies in AI-mediated bilingual classrooms.

2.1 CLIL 4Cs framework

The CLIL 4Cs remain central to bilingual education. They articulate the pedagogical balance between subject knowledge, language use, higher-order thinking, and intercultural awareness (Coyle, 2007). However, in AI-mediated contexts, these four dimensions require reinterpretation. For example, *communication* now involves interactions with AI tools such as chatbots, while *cognition* may be supported or disrupted by AI-generated feedback.

2.2 TPACK framework

The TPACK framework (Mishra and Koehler, 2006) conceptualizes teacher knowledge as the integration of technological, pedagogical, and content knowledge. While widely applied in digital education research, TPACK does not fully capture the complexities of AI-mediated learning environments (Khalil and Alsenaidi, 2024; Korbek and Wolf, 2020).

First, TPACK treats technology primarily as a tool, whereas generative AI increasingly functions as an active agent in knowledge production. Second, it does not explicitly address AI literacy, including the ability to critically evaluate AI-generated outputs. Third, it lacks a systematic treatment of digital ethics, such as bias, transparency, and academic integrity. Finally, TPACK does not account for bilingual instructional contexts such as CLIL.

In response, the CAIIL-TC framework extends TPACK by incorporating AI literacy, ethical reasoning, and AI-mediated bilingual pedagogy, thereby providing a more comprehensive model for pre-service teacher competencies.

2.3 AI literacy

AI Literacy emphasizes the ability to understand, use, and critically evaluate AI (Long and Magerko, 2020; Zeevy Solovey, 2024). In teacher education, this includes competencies such as prompt engineering, awareness of algorithmic bias, and the

capacity to adapt AI for student-centered learning. AI Literacy thus directly complements TPACK: where TPACK describes knowledge domains, AI Literacy equips teachers with strategies to act within those domains in AI-mediated environments.

2.4 Digital ethics

Digital Ethics foregrounds the values and responsibilities guiding technology use in education, including transparency, fairness, data privacy, and accountability (Livingstone and Sefton-Green, 2016). In CLIL classrooms, ethical considerations become more complex when AI-generated content intersects with issues of academic integrity, linguistic representation, and cultural sensitivity. Ethical awareness thus intersects with the *culture* dimension of the 4Cs, ensuring that AI-mediated teaching respects diversity and promotes inclusion.

2.5 Intersections across frameworks

Rather than viewing these frameworks as separate silos, this review highlights their interactions:

1. CLIL 4Cs × TPACK: AI reshapes how content and language are jointly taught, requiring new task designs that balance linguistic scaffolding with disciplinary rigor.
2. TPACK×AI Literacy: Teachers need not only knowledge of technology integration but also the ability to critically design and adapt AI-mediated activities.
3. AI Literacy×Digital Ethics: Competence in prompt design must be accompanied by ethical awareness to avoid reinforcing bias or misuse.
4. CLIL 4Cs × Digital Ethics: Intercultural learning requires careful curation of AI-generated content to avoid stereotyping or misinformation.

3 Literature review approach

This study adopts a theoretical review approach (Grant and Booth, 2009), which is appropriate for synthesizing conceptual frameworks and generating new theoretical insights. The aim is not to aggregate effect sizes, but to consolidate and extend existing theoretical perspectives to inform the development of the CAIIL-TC framework for AI-mediated CLIL teacher education.

3.1 Research questions

This review is guided by three research questions:

1. What competencies for AI-mediated CLIL instruction are identified in the literature?
2. How do CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics interact?
3. How can these be synthesized into a new competency framework?

3.2 Search strategy

The literature search was conducted using Web of Science, Scopus, and ERIC, covering publications from 2010 to 2025. Search terms included combinations of:

- ("AI in education" OR "artificial intelligence" OR "AI literacy") AND ("teacher education" OR "pre-service teacher")
- ("CLIL" OR "content and language integrated learning") AND ("competency" OR "framework")
- ("TPACK") AND ("AI" OR "technology integration")
- ("digital ethics" OR "algorithmic bias") AND ("education")

Following this procedure, a total of 312 records were identified. After removing duplicates, 256 articles remained for screening. Based on title and abstract relevance, 118 articles were selected for full-text review. After applying inclusion and exclusion criteria, 63 studies were retained, of which 38 studies directly informed the development of the conceptual framework.

The selection process is summarized in Figure 1.

3.3 Inclusion and exclusion criteria

Inclusion criteria:

- Peer-reviewed journal articles, book chapters, or policy reports
- Published in English
- Relevant to AI, CLIL, or teacher competencies

Exclusion criteria:

- Purely technical AI studies without pedagogical relevance
- Non-scholarly commentary

3.4 Analytical procedure

The analysis was conducted in two stages. First, studies were coded based on alignment with four frameworks: CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics. Second, a cross-framework synthesis identified overlaps, complementarities, and gaps.

The coding process was iterative, with categories refined through repeated comparison across studies to ensure consistency. To reduce interpretive bias, a systematic cross-checking approach was applied between conceptual categories and source materials.

3.5 Rationale

A theoretical review is appropriate given the emerging nature of AI-mediated CLIL. This approach enables conceptual integration across fragmented research and supports the development of a new framework for future empirical validation.

To enhance transparency, a summary of representative studies included in the synthesis is presented in Table 1. These studies were selected to reflect key thematic areas across AI in education, CLIL pedagogy, AI literacy, and digital ethics. Rather

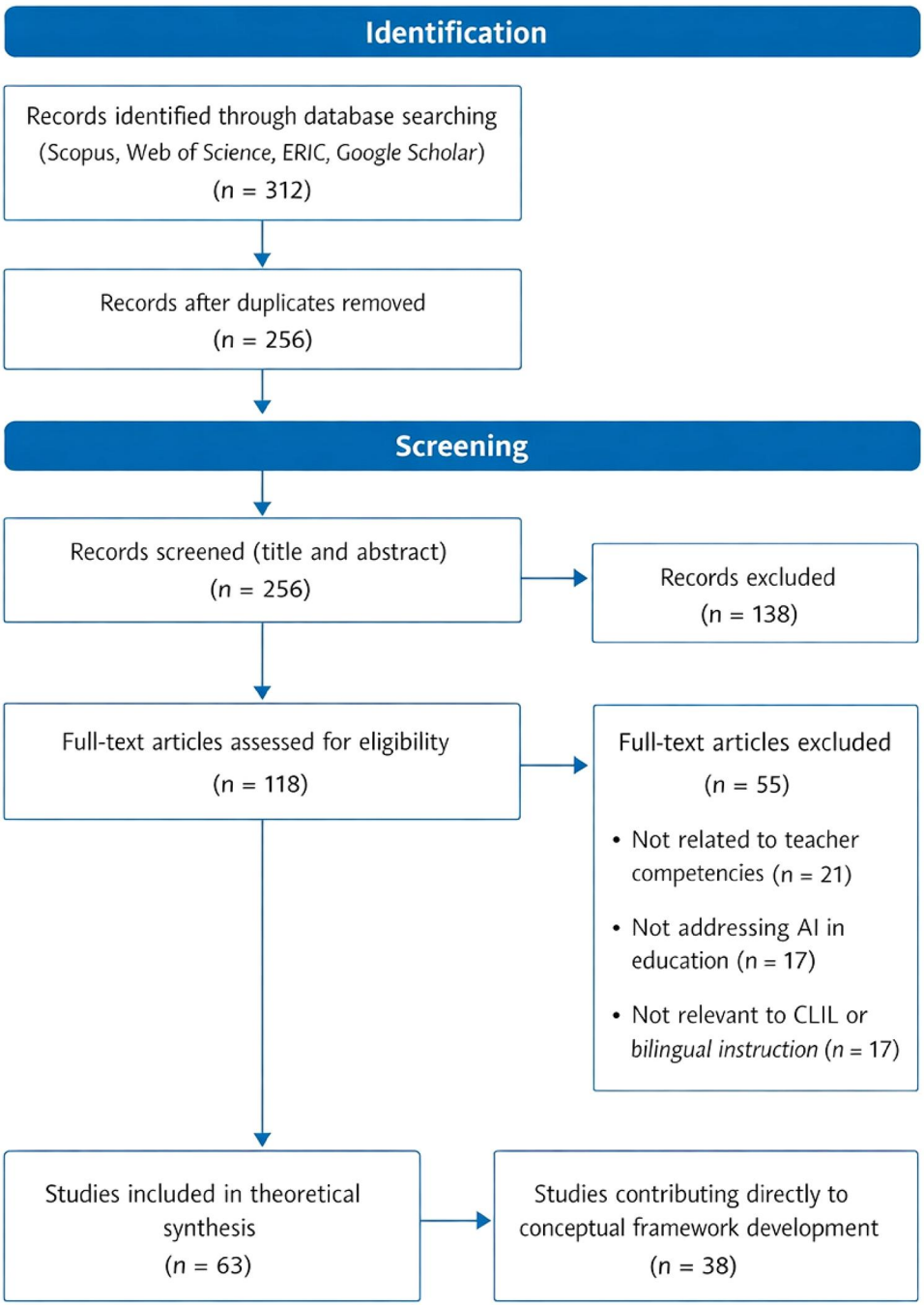


FIGURE 1
PRISMA flow diagram of literature selection.

than providing an exhaustive list, the table highlights influential works that directly informed the development of the CAIIL-TC competency domains.

3.6 Qualitative coding procedure

The qualitative analysis employed a systematic, theory-driven three-stage coding process: open, axial, and selective coding. Open coding identified concepts related to AI-mediated CLIL competencies; axial coding organized these into broader categories linking pedagogical functions, ethical considerations, and competency domains; selective coding synthesized core themes into the CAIIL-TC framework. Two independent coders with expertise in educational technology and CLIL ensured consistency using an iterative codebook. Inter-coder reliability was substantial (Cohen's $\kappa = 0.82$). The complete codebook is provided in Appendix A.

4. Development of the CAIIL-TC Framework

The CAIIL-TC framework (Competencies for AI-Integrated CLIL Teacher Education) was developed to address the gaps

TABLE 1 Representative studies included in the literature synthesis.

Author(s)	Year	Focus area	Method/type	Key contribution to CAIIL-TC
Zawacki-Richter et al.	2019	AI in Higher Education	Systematic Review	Identifies gaps in pedagogical integration of AI; supports need for AI Content Evaluation
Kasneci et al.	2023	Generative AI in Education	Conceptual Analysis	Highlights risks of LLMs (bias, hallucination); informs Ethical & Reflective Use
Crompton and Burke	2023	AI in Higher Education	Systematic Review	Emphasizes teacher mediation role; supports Collaborative Mediation
Holmes et al.	2022	AI & Education Policy	Conceptual/Policy	Establishes ethical frameworks; informs Ethical & Reflective Use
Floridi et al.	2018	AI Ethics	Theoretical Framework	Defines fairness, accountability; supports Digital Ethics dimension
Lee and Song	2024	AI-generated learning	Empirical Study	Shows limitations of AI explanations; informs Generative Input Design
Park and Kim	2023	AI interaction design	Empirical Study	Explores human-AI interaction; supports task design competencies
Kohnke	2024	GenAI in Language Learning	Empirical Study	Highlights need for adaptation; informs Contextual Re-Authoring
Guan et al.	2025	Pre-service teachers & AI	Empirical Study	Shows gaps in teacher readiness; supports framework necessity
Alexandrowicz	2024	AI in teacher education	Review	Identifies cognitive and pedagogical challenges; supports all domains
Son et al.	2025	AI in language learning	Review	Demonstrates AI affordances in language pedagogy; informs CLIL integration
Long and Magerko	2020	AI Literacy	Conceptual	Defines AI literacy competencies; supports all AI-related domains
UNESCO	2021	AI policy in education	Policy Report	Establishes global AI literacy framework; supports institutional relevance

identified in the literature review and to provide a structured competency model for pre-service teacher preparation. This framework synthesizes insights from CLIL pedagogy, TPACK, AI Literacy, and Digital Ethics, while extending them into domains that specifically address the challenges and opportunities of AI-mediated bilingual instruction.

4.1 Linking literature to framework

The development of the CAIIL-TC framework followed an iterative synthesis of the reviewed literature. Studies were analyzed for their relevance to teacher competencies in AI-mediated education, and recurrent themes were clustered into competency domains. Each domain of CAIIL-TC thus reflects both the conceptual foundations of existing frameworks and empirical findings from recent studies.

1. AI Content Evaluation: This competency reflects teachers' ability to critically assess AI-generated texts, translations, and learning materials for accuracy, bias, and appropriateness. Prior research highlights the risks of misinformation and cultural insensitivity in AI outputs (Kasneci et al., 2023; Zawacki-Richter et al., 2019). Within CLIL contexts, this aligns with the *content* and *culture* dimensions of the 4Cs, ensuring that AI-mediated materials support disciplinary rigor and intercultural sensitivity.
2. Generative Input Design: Building on research in prompt engineering and AI-assisted task design (Lee and Song, 2024; Park and Kim, 2023), this competency requires teachers to craft effective inputs that elicit pedagogically valuable outputs from AI. It extends TPACK's focus on technology use by incorporating AI Literacy skills, highlighting the need for creativity and intentionality in AI-mediated CLIL task design.
3. Contextual Re-Authoring: This domain reflects the teacher's role in adapting or re-authoring AI outputs to align with curricular goals and student needs. Studies of ChatGPT in writing instruction show that unedited AI responses often lack contextual relevance (Kohnke, 2024). In CLIL classrooms, re-authoring ensures that AI-generated texts are linguistically scaffolded and culturally responsive, reinforcing both the *pedagogical* and *cultural* dimensions of teacher competence.
4. Ethical and Reflective Use: Echoing the literature on digital ethics and academic integrity (Floridi et al, 2018; Holmes et al., 2022), this competency emphasizes transparency, fairness, and accountability in AI use. Teachers must be able to model ethical practices for students, such as citing AI-assisted content and recognizing algorithmic bias. This aligns with the *culture* dimension of CLIL and the ethical strand of AI Literacy.
5. Collaborative Mediation: Research on teacher–student interactions in AI-mediated environments (Crompton and Burke, 2023) highlights the importance of collaborative meaning-making. Teachers must act as mediators between AI tools and learners, ensuring that AI functions as a complement to, rather than a replacement for, human interaction. This competency resonates with TPACK's pedagogical dimension and CLIL's emphasis on communication.

4.2 Structure of the CAIIL-TC framework

The CAIIL-TC framework is conceptualized as an integrative and action-oriented competency model that captures how pre-service teachers enact AI-mediated CLIL instruction in practice. Rather than treating competencies as discrete knowledge domains, the framework emphasizes their dynamic interaction and situated enactment within bilingual learning environments supported by AI technologies.

The framework comprises five interrelated domains: AI Content Evaluation, Generative Input Design, Contextual Re-Authoring, Ethical and Reflective Use, and Collaborative Mediation. These domains represent not only areas of professional knowledge but also pedagogically embedded practices that emerge when teachers engage with AI tools in CLIL contexts. In this respect, CAIIL-TC extends beyond static competency classifications and reconceptualizes teacher expertise as a form of adaptive and context-responsive performance.

Structurally, the framework can be understood through three complementary dimensions:

First, a pedagogical process dimension.

The five domains reflect a typical instructional workflow in AI-mediated CLIL classrooms. Teachers begin by evaluating AI-generated materials (AI Content Evaluation), followed by designing prompts or tasks to generate pedagogically meaningful outputs (Generative Input Design). These outputs are then adapted to align with learners' linguistic proficiency, disciplinary objectives, and sociocultural contexts (Contextual Re-Authoring). Throughout this process, Ethical and Reflective Use functions as a guiding principle, ensuring responsible, transparent, and pedagogically appropriate engagement with AI. Finally, Collaborative Mediation positions teaching as an interactive process in which teachers orchestrate meaningful engagement between learners and AI systems.

Second, a cross-framework integration dimension.

Each domain draws upon, but also extends, elements from CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics. Importantly, CAIIL-TC does not merely combine these frameworks; it reconfigures their intersections to address the specific affordances and constraints of generative AI in bilingual education. For example, Generative Input Design expands TPACK by incorporating AI-specific task construction practices, while Ethical and Reflective Use embeds ethical reasoning directly into pedagogical decision-making rather than positioning it as an external consideration.

Third, an interdependent competency system.

The domains function as a mutually reinforcing system rather than a linear sequence. Effective Generative Input Design presupposes competence in AI Content Evaluation, while Contextual Re-Authoring requires the integration of pedagogical knowledge and cultural awareness. Similarly, Collaborative Mediation depends on the alignment of ethical judgment, technological understanding, and communicative competence. This interdependence reflects the complexity of real-world teaching, where multiple competencies are enacted simultaneously in response to evolving classroom conditions.

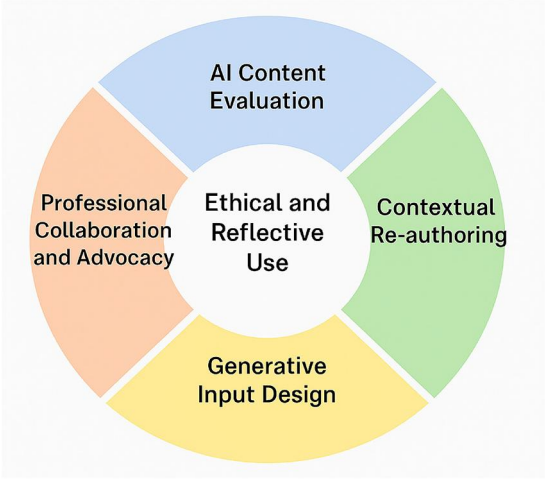


FIGURE 2
CAIIL-TC competency framework for AI-mediated CLIL instruction.

The structural relationships among these domains are illustrated in Figure 2, which presents the CAIIL-TC framework as an integrated model of AI-mediated CLIL competencies. The analytical alignment of these domains with existing frameworks and their distinct contributions are further elaborated in Section 7.

5 Challenges in current practice

Despite the growing emphasis on artificial intelligence (AI) in education and the pedagogical merits of Content and Language Integrated Learning (CLIL), the integration of AI into pre-service bilingual teacher education remains fraught with challenges. These challenges are not merely technical but also institutional, cognitive, and cultural. Understanding these barriers is crucial for the practical implementation of any AI-integrated CLIL competency model, including CAIIL-TC.

5.1 Institutional inertia and fragmented policy support

In Taiwan, pre-service teacher education is still largely governed by traditional curriculum frameworks that separate subject teaching, language pedagogy, and digital competence. AI integration is often treated as a general trend in EdTech rather than a component of language and content instruction. Although national initiatives such as the "Digital Learning White Paper" (Ministry of Education Taiwan, 2022) have introduced AI literacy into K–12 education, teacher education institutions have yet to embed AI into CLIL-specific courses.

For instance, the national teacher certification exam for secondary English and subject teachers does not assess candidates' ability to integrate AI tools or evaluate AI-generated learning content. This mismatch limits incentives for both teacher educators and student teachers to prioritize AI competencies. In contrast, some European countries such

as Finland and Estonia have begun including digital pedagogical reasoning and data ethics in their national teacher education curricula (European Commission, 2023; Redecker, 2017), with localized adaptations for CLIL programs in multilingual regions.

5.2 Cognitive load and competency fragmentation

Pre-service teachers are expected to master complex instructional practices involving both language and content pedagogy, often in their non-dominant language. Adding AI into this already demanding context introduces what many perceive as "cognitive overload" (Alexandrowicz, 2024). A recent observation in Taiwanese teacher education programs suggests that many student teachers perceive AI tool integration as confusing and difficult to align with curricular goals, especially in science and social studies CLIL units. This aligns with Hwang and Chien's (2022) findings, which highlight that many educators in Taiwan report insufficient preparation and pedagogical guidance when integrating AI tools into their teaching.

Moreover, training on AI in teacher education programs is often delivered as standalone workshops, disconnected from the bilingual practicum or language education courses. This fragmented approach prevents student teachers from developing contextualized, pedagogically sound uses of AI, leading to reliance on superficial applications such as AI-generated translations or automated quiz creation, without critical reflection on appropriateness or language levels (Alsaedi, 2024; Meniado et al., 2024).

5.3 Cultural tensions and resistance to innovation

Teacher identity and professional culture also contribute to resistance. In Taiwanese contexts, both pre-service and mentor teachers express concerns about AI weakening their role as expert communicators (Beauchamp and Thomas, 2009), particularly in language teaching. Recent studies have revealed that many pre-service teachers feel inadequately prepared to implement AI-integrated pedagogy, citing low confidence in their digital capabilities and uncertainty about how AI aligns with their teacher identity (Guan et al., 2025). In Europe, CLIL teachers in bilingual regions like Catalonia and South Tyrol report similar concerns, where teachers emphasize human interaction as central to translanguaging and identity negotiation in the classroom (Cenoz and Gorter, 2017).

Additionally, pre-service teachers often struggle to balance innovation with accountability, fearing that experimental use of AI in teaching practices may not be welcomed by conservative mentor teachers or school administrators. This results in cautious minimal use of AI tools, even when teacher candidates express high interest in leveraging AI for content-rich language tasks.

6 Comparative analysis and recommendations

The integration of AI into teacher education is unfolding unevenly across regions, shaped by policy priorities, institutional structures, and cultural contexts (McNulty, 2025). While both Europe and Asia have made progress, comparative analysis reveals differences in implementation depth and systemic challenges. This section deepens the examination of Finland and Taiwan as representative cases and extends the scope to include emerging initiatives in Latin America and the Middle East. To enhance cross-regional interpretability, a structured comparative synthesis is provided in Table 2.

6.1 Finland: institutionalization of AI literacy

Finland has been widely recognized for its Elements of AI initiative, a free online course designed to introduce AI concepts to the general public. While the course has reached a broad audience, its integration into teacher education has been more complex. Recent studies show that Finnish universities of education have selectively embedded Elements of AI into digital pedagogy modules, yet pre-service teachers often report challenges in connecting abstract AI concepts to classroom practices (Shahzad et al., 2025). Moreover, institutional uptake varies, with some faculties treating AI literacy as optional enrichment rather than a core requirement. These findings suggest that while Finland leads in national AI awareness, its translation into teacher competencies remains partial and uneven. This reflects a broader pattern identified in the coding analysis, where "AI_USE" is present but weakly connected to "LANG_SCAF" and "CONTENT_LEARN" in pedagogical practice.

TABLE 2 Comparative overview of AI integration in pre-service teacher education.

Dimension	Finland	Taiwan	Latin America	Middle East
Policy Orientation	National AI literacy initiatives	Bilingual policy with AI support	Emerging digital education reforms	National AI strategies
Implementation Level	Partial institutional integration	Policy-driven but uneven	Pilot programs	Advanced in elite institutions
Teacher Preparedness	Moderate	Limited AI pedagogical training	Emerging	Uneven
Key Challenge	Theory-practice gap	Institutional resistance	Resource constraints	Institutional disparity
Pedagogical Focus	General AI literacy	AI-integrated CLIL	Data literacy, ethics	AI literacy modules

6.2 Taiwan: policy ambition and institutional resistance

Taiwan's bilingual education policy, launched in 2018, has positioned AI as a tool for advancing CLIL instruction in schools. However, implementation has faced several systemic obstacles. Teacher surveys reveal concerns about the alignment of AI-assisted materials with curricular goals (Chen et al., 2025), while institutional resistance has slowed the adoption of AI in teacher preparation programs. Faculty development opportunities remain limited, and pre-service teachers often lack training in AI literacy beyond basic technology integration workshops. These gaps underscore the risk of policies being top-down without sufficient institutional support and bottom-up capacity building. Consistent with the CAIIL-TC framework, this case highlights the role of "technology anxiety" and limited "technology acceptance" in constraining effective AI-integrated CLIL implementation.

6.3 Latin America: emerging experiments

In Latin America, AI initiatives in education are more fragmented but offer important insights. In Brazil, AI has been piloted in teacher training programs for mathematics and science, emphasizing data literacy and ethical awareness (Rivas, 2025). Chile has launched AI literacy workshops for pre-service teachers as part of broader digital transformation projects. While these initiatives remain small in scale, they demonstrate growing recognition of AI as an essential component of teacher professional development, particularly in resource-constrained contexts. From a coding perspective, this region shows emerging alignment between "TECH_ACCEPT" and "AI_USE," though institutional scalability remains limited.

6.4 Middle East: strategic investment

The United Arab Emirates (UAE) has positioned AI as a strategic national priority. The establishment of the Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) reflects this ambition, and teacher education programs are beginning to incorporate AI literacy modules into pre-service curricula (EIN Presswire, 2025). However, similar to Finland, implementation across institutions remains uneven, with elite universities more advanced than regional teacher colleges. Ethical concerns regarding data governance and language diversity are also salient in the Middle Eastern context. This reflects a dual pattern identified in the analysis: strong "AI_USE" presence but uneven distribution across institutional contexts, raising concerns related to "AI_DEP" and ethical considerations.

6.5 AI-mediated CLIL implementation scenario

To enhance practical relevance, an AI-mediated CLIL classroom scenario is introduced to illustrate how AI-mediated CLIL can be operationalized in teacher education.

In a secondary-level science CLIL classroom, students engage in a unit on climate change and ocean ecosystems. The teacher integrates an AI conversational system to scaffold both content understanding and language production.

1. Pre-task: AI supports vocabulary acquisition and sentence construction
2. During-task: Students use AI to interpret complex scientific concepts and generate summaries
3. Post-task: AI prompts reflection on key scientific concepts and reinforces language use

This scenario demonstrates how AI-mediated CLIL supports both CONTENT_LEARN and LANG_SCAF, while also highlighting potential risks of AI dependency, such as overreliance on AI-generated outputs.

6.6 Recommendations for pre-service teacher education

From this comparative analysis, several recommendations emerge:

1. Institutionalize AI literacy: AI training should move from optional enrichment to core components of pre-service teacher curricula.
2. Contextualize AI in CLIL practice: Teacher education must link AI tools to subject-specific and bilingual teaching tasks, rather than treating them as generic technologies.
3. Invest in faculty development: Teacher educators require ongoing support to model AI-integrated pedagogy, ensuring sustainability of reforms.
4. Address ethical and cultural dimensions: Regional implementations must foreground digital ethics and cultural inclusivity, especially when AI-generated content is used in linguistically and culturally diverse classrooms.
5. Encourage cross-regional exchange: Programs should draw lessons not only from Europe and Asia but also from emerging cases in Latin America and the Middle East, fostering global perspectives.

7 Proposed competency framework

The CAIIL-TC framework synthesizes insights from the literature and extends existing models to articulate the competencies required for pre-service teachers engaging in AI-mediated CLIL instruction. Building upon the foundations of CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics, CAIIL-TC identifies five interrelated domains: AI Content Evaluation, Generative Input Design, Contextual Re-Authoring, Ethical and Reflective Use, and Collaborative Mediation.

7.1 Framework matrix

To illustrate the alignment between these domains and existing teacher education standards, a single integrative matrix is presented below (Table 3). This matrix maps CAIIL-TC

TABLE 3 Mapping of CAIIL-TC competencies against existing frameworks.

CAIIL-TC competency	CLIL 4Cs	TPACK	AI literacy	Digital ethics	Distinct contribution of CAIIL-TC
AI Content Evaluation	Content, Culture	Content Knowledge	Understanding AI outputs	Fairness, accuracy, bias awareness	Introduces critical evaluation of AI-generated knowledge in CLIL contexts
Generative Input Design	Cognition, Communication	Technological Pedagogical Knowledge	Prompt engineering, AI task design	Responsible adaptation of AI	Extends TPACK to include generative AI task design
Contextual Re-Authoring	Communication, Culture	Pedagogical Knowledge	Adapting AI-generated content	Transparency, inclusivity	Emphasizes teacher role as editor and contextual mediator
Ethical and Reflective Use	Culture	Pedagogical Knowledge	Awareness of bias, responsible use	Accountability, academic integrity	Positions ethics as central to AI-mediated pedagogy
Collaborative Mediation	Communication, Cognition	Technological Pedagogical Knowledge	Human–AI interaction	Human-centered AI use	Defines teacher–AI–student co-agency in learning

competencies onto representative dimensions of the four theoretical frameworks. It clarifies how CAIIL-TC both builds on established knowledge and introduces novel elements specific to AI-mediated bilingual education.

This integrative view demonstrates that while CAIIL-TC shares intersections with prior frameworks, it extends them in significant ways. For example, *Generative Input Design* goes beyond TPACK's focus on technology knowledge by incorporating AI-specific task design, while *Contextual Re-Authoring* and *Ethical and Reflective Use* add competencies not fully represented in existing models.

7.2 Interrelated competencies

Importantly, the five competencies should not be seen as isolated domains. They operate dynamically in classroom practice: effective *Generative Input Design* presupposes competence in *AI Content Evaluation*, while *Collaborative Mediation* requires teachers to combine ethical awareness with pedagogical adaptability. This interconnectedness reflects the complex demands of AI-mediated CLIL and underscores the need for holistic preparation in teacher education.

7.3 Implications for teacher education

By consolidating insights into a single framework, CAIIL-TC provides a conceptual foundation for curriculum design in pre-service teacher education. It offers a structured yet flexible guide that can inform course development, practicum supervision, and professional standards. In addition, the framework points toward areas where empirical research is needed to validate and refine the proposed competencies.

To further enhance applicability, the framework can be implemented through structured teacher education modules that integrate AI tools into CLIL lesson design, assessment, and classroom interaction. Comparative insights from different regions suggest that flexibility is essential, allowing institutions to adapt the framework to local policy contexts, technological infrastructure, and teacher readiness.

8 Conclusion and future directions

This article has proposed the CAIIL-TC framework (Competencies for AI-Integrated CLIL Teacher Education) to address the urgent need for conceptual guidance in preparing pre-service teachers for AI-mediated bilingual classrooms. By synthesizing four influential theoretical models(CLIL 4Cs, TPACK, AI Literacy, and Digital Ethics), this framework identifies five interrelated competency domains: AI Content Evaluation, Generative Input Design, Contextual Re-Authoring, Ethical and Reflective Use, and Collaborative Mediation.

8.1 Theoretical contributions

The CAIIL-TC framework makes three key contributions:

1. Integration of frameworks: It bridges well-established models (CLIL 4Cs, TPACK) with emerging perspectives (AI Literacy, Digital Ethics), offering a more comprehensive understanding of teacher competencies in the age of AI.
2. Extension beyond existing models: While TPACK captures knowledge integration and the 4Cs emphasize bilingual pedagogy, CAIIL-TC introduces new domains—such as *Generative Input Design* and *Contextual Re-Authoring*—that directly respond to the affordances and challenges of generative AI.
3. Conceptual foundation for practice: CAIIL-TC provides a structured lens through which pre-service teacher curricula can be redesigned, ensuring that future educators are equipped not only with knowledge but also with ethical and adaptive competencies for AI-integrated classrooms.

8.2 Limitations

As a theoretical review, this study did not conduct empirical validation of the proposed framework. While the CAIIL-TC model is grounded in literature synthesis, its practical applicability remains to be tested. Additionally, regional comparisons included Finland, Taiwan, and selected cases from Latin America and the Middle East, but future studies should broaden the scope to other regions to capture a more global perspective. Future research may also develop and test specific classroom implementation scenarios to further validate the practical applicability of the framework.

8.3 Future research directions

To strengthen and validate the CAIIL-TC framework, we recommend several avenues for future research:

1. Empirical validation: Conduct classroom-based studies to examine how pre-service teachers engage with each competency domain in practice, using both qualitative and quantitative methods.
2. Longitudinal studies: Track teacher development across teacher education programs to assess how competencies evolve over time, particularly as AI tools become more embedded in pedagogy.
3. Cross-cultural comparisons: Compare how AI-integrated CLIL competencies manifest across diverse educational systems, including underrepresented regions such as Africa and South Asia.
4. Instrument development: Design and validate assessment tools or self-report instruments that measure teachers' competencies in CAIIL-TC domains.
5. Policy and curriculum design: Explore how the framework can inform national teacher standards and accreditation systems, ensuring alignment between conceptual models and institutional practices.

8.4 Closing remarks

In conclusion, the CAIIL-TC framework provides a timely and innovative contribution to the discourse on teacher education in the AI era. While conceptual at this stage, it establishes a strong foundation for future empirical work. By connecting theoretical insights with practical imperatives, the framework invites teacher educators, policymakers, and researchers to collaboratively advance the preparation of future teachers for the complex realities of AI-mediated CLIL instruction.

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The author(s) declared that generative AI was not used in the creation of this manuscript.

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Appendix A. Codebook

A1. Coding Framework Overview

The codebook was developed through an iterative process during open, axial, and selective coding. It includes code definitions, inclusion and exclusion criteria, and representative examples to ensure consistency and transparency in the coding process.

A2. Primary Codes and Definitions

Code	Definition	Inclusion Criteria	Exclusion Criteria	Example
AI_USE	Use of AI tools in teaching and learning contexts	AI-generated content, AI-assisted instruction, conversational AI use	General ICT tools without AI features	Use of AI chatbot to explain concepts
CONTENT_LEARN	Subject-matter learning within CLIL contexts	Science, mathematics, or other disciplinary knowledge	Language-only activities	Understanding ocean acidification
LANG_SCAF	Language scaffolding to support comprehension and production	Vocabulary support, sentence construction, paraphrasing	Pure content explanation without language focus	AI suggests sentence structures
AI_EVAL	Evaluation of AI-generated outputs	Checking accuracy, bias, or relevance of AI content	Passive use of AI outputs	Teacher verifies correctness of AI explanation
GEN_INPUT	Design of AI-generated instructional input	Prompt design, task creation using AI	General lesson planning without AI	Teacher creates AI prompts for tasks
RE_AUTHOR	Adaptation and modification of AI-generated content	Editing, contextualizing, simplifying AI outputs	Direct copy-paste from AI	Teacher modifies AI-generated text
ETH_USE	Ethical and responsible use of AI	Academic integrity, bias awareness, responsible use	Technical use without ethical consideration	Discussion of AI-generated bias
COL_MED	Collaborative mediation among teacher, student, and AI	Interaction involving human-AI collaboration	Individual AI use without interaction	Teacher guides student-AI interaction
AI_DEP	Over-reliance on AI in learning or teaching	Dependence on AI for answers or task completion	Balanced or guided AI use	Student relies solely on AI-generated responses

A3. Axial Categories

Category	Related Codes
Pedagogical Functions	AI_USE, CONTENT_LEARN, LANG_SCAF
Instructional Design	GEN_INPUT, RE_AUTHOR
Evaluation and Ethics	AI_EVAL, ETH_USE
Interactional Processes	COL_MED
Risk Factors	AI_DEP

A4. Selective Coding

The final stage of coding resulted in the identification of the following core themes:

1. Integration of AI in CLIL pedagogy requires both instructional design and evaluative competencies
2. Teacher roles extend from knowledge transmission to mediation and adaptation of AI-generated content
3. Ethical use and critical evaluation are essential components of AI-mediated teaching
4. AI dependency represents a potential risk in AI-mediated learning environments

A5. Reliability and Coding Notes

1. Coding was conducted by two independent coders
2. Code definitions were refined iteratively during the coding process
3. Ambiguous cases were resolved through discussion and consensus
4. Inter-coder reliability reached Cohen's $\kappa = 0.82$, indicating substantial agreement